

# Inter-Annotator Agreement Analysis Across Error Categories

## 1. Composite Consistency Score

For each combination of rater group, error category, and evaluation task, all available IAA metric values are collected and averaged to produce a **composite consistency score**. Binary tasks, Detection (C1) and Category (C3), contribute up to five metrics: Fleiss’  $\kappa$ , Gwet’s  $AC_1$ , Krippendorff’s  $\alpha$  (nominal), mean pairwise Cohen’s  $\kappa$ , and mean Spearman  $\rho$ . Ordinal tasks, Span + Description (C2) and Correction (C4), add Gwet’s  $AC_2$ , Krippendorff’s  $\alpha$  (ordinal), and ICC(A,1), giving up to eight values per cell. Undefined values (e.g. Spearman  $\rho$  when one rater’s scores carry zero variance) are excluded; the mean is computed over available values only. Per-task means are then averaged across the four tasks to yield one composite per (group, category) pair. The composite inherits the  $[-1, +1]$  scale of the underlying metrics; values at or below zero indicate sub-chance agreement.

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# Step 1: collect all valid metric values for one (group, category, task)
metric_values = [fleiss_k, gwet_ac1, kripp_alpha, cohen_k, spearman_rho]
if task in ["C2", "C4"]: # ordinal tasks add three more
    metric_values += [gwet_ac2, kripp_alpha_ord, icc_a1]

task_mean = np.mean([v for v in metric_values if not np.isnan(v)])

# Step 2: average the four task means -> composite
composite = np.mean([task_mean_C1, task_mean_C2, task_mean_C3, task_mean_C4])
```

Table 1: Composite consistency scores by rater group and error category.

| Error Category                                                                                                        | G1H-H        | G2Ann-Ann    | G3NonAnn-NonAnn | G5H+Ann      | G6H+NonAnn   | G7All        |
|-----------------------------------------------------------------------------------------------------------------------|--------------|--------------|-----------------|--------------|--------------|--------------|
| Script Norm.                                                                                                          | 0.854        | -0.043       | 0.639           | 0.088        | 0.216        | 0.170        |
| Spelling                                                                                                              | 0.902        | 0.065        | 0.636           | 0.161        | <b>0.540</b> | 0.310        |
| Grammatical                                                                                                           | 0.885        | 0.030        | 0.399           | 0.183        | 0.457        | 0.265        |
| Code-Mixing                                                                                                           | <b>0.962</b> | <b>0.140</b> | 0.471           | <b>0.293</b> | 0.529        | <b>0.392</b> |
| Correct                                                                                                               | 0.839        | -0.039       | 0.390           | 0.372        | 0.332        | 0.300        |
| <i>Mean across groups</i> Code-Mixing 0.464 > Spelling 0.436 > Grammatical 0.370 > Correct 0.366 > Script Norm. 0.321 |              |              |                 |              |              |              |

## 2. Overall Category Ranking

Averaged across all six rater groups, **Code-Mixing / Wrong Language** yields the highest composite (0.464, mean rank 1.67), followed by Spelling and Typographical Error (0.436). Script Normalization is the least consistently evaluated category (0.321, mean rank 4.17). The ordering is stable across groups: Code-Mixing ranks first or second in all six groups, and Script Normalization ranks fourth or fifth in five of the six. This cross-group stability indicates that the difficulty ordering is a property of the error types rather than of individual evaluator behaviour.

### 3. Group-by-Group Analysis

**G1 (Human–Human).** All five categories yield high composite scores (0.839–0.962), confirming reliable human agreement across every error type. Code-Mixing is easiest (0.962): Roman-script/Non-Odia tokens in an Odia sentence constitute an unambiguous script-level signal. Correct Sentence / No Errors is hardest (0.839), reflecting residual disagreement on whether syntactically marginal sentences are truly error-free. Grammatical Error ranks third (0.885), indicating that native-speaker expertise is sufficient for strong morpho-syntactic agreement.

**G2 (Annotator LLM–Annotator LLM).** Composite agreement scores drop to nearly zero or lower (−0.043 to 0.140). The conflict of interest inherent in self-evaluation causes Sarvam-105B and LLaMA-4-Scout-17B to consistently disagree on which system performed better, regardless of the error category. Agreement for Script Normalization (−0.043) and Correct Sentence (−0.039) falls below the levels expected from random chance. Only Code-Mixing maintains a positive score (0.140), as the clear visual signal of switching between different scripts is strong enough to partly overcome the models’ self-preference bias.

**G3 (Non-Annotator LLM–Non-Annotator LLM).** Composites are moderate (0.390–0.639), substantially above G2, confirming that removing the self-interest conflict restores meaningful agreement. Script Normalization ranks first here (0.639), reversing its standing in all other groups. Script-variant errors are detectable via pattern matching, which these multilingual LLMs can perform without requiring deep Odia morphological knowledge. Grammatical Error ranks fourth (0.399), indicating greater divergence on syntactic errors than on typographical or cross-script ones.

**G5 (Humans + Annotator LLMs).** Group agreement is markedly reduced (0.088–0.372) by the inclusion of self-evaluating annotator LLMs. Correct Sentence ranks first (0.372). Script Normalization falls to 0.088, the lowest cell in the table outside G2.

**G6 (Humans + Non-Annotator LLMs).** This is the strongest mixed group. Spelling ranks first (0.540), followed by Code-Mixing (0.529) and Grammatical Error (0.457). The combination of human expertise and calibrated third-party LLMs preserves the human gold-standard signal while broadening evaluative coverage. Grammatical Error reaching 0.457 (third highest in the group) provides the clearest evidence that grammatical judgements, though demanding, are accessible to capable non-annotator LLMs.

**G7 (All Raters).** Including the annotator LLMs reduces all scores below G6 levels. Code-Mixing remains first (0.392) and Script Normalization last (0.170). The ordinal category ranking is preserved despite the lower absolute values, confirming the robustness of the difficulty ordering.

### 4. Cross-Cutting Patterns

Three patterns emerge consistently across groups.

*Code-Mixing* ranks first or second in all six groups. The presence of Non-Odia tokens in an Odia sentence is a script-level surface signal requiring no grammatical knowledge to detect, making this the most evaluator-agnostic category.

*Script Normalization* is the only category that reverses ranking between group types: it is the hardest for human and human-inclusive groups but the easiest for LLM-only groups (G3), which exploit pattern matching over character variants rather than linguistic reasoning. This evaluator-type dependency means Script Normalization consistency values are not directly comparable across human and LLM groups.

*Grammatical Error* occupies the middle of the ranking (3rd or 4th) across all groups. C3 agreement is relatively high, indicating that the error category label is identifiable. C4 agreement is the lowest

of all tasks for this category, reflecting the additional demand of generating a morpho-syntactically correct alternative form.

**Appendix: Per-Metric IAA Tables by Error Category and Rater Group**

Table 2: Fleiss’  $\kappa$  / Gwet’s  $AC_1$  (binary tasks C1, C3) and Gwet’s  $AC_2$  (ordinal tasks C2, C4) by error category and rater group. Fleiss’  $\kappa = AC_1$  exactly for binary tasks.

| Rater Group                 | Error Category | $C1_{\kappa_F} = AC_1$ | $C2AC_1 / AC_2$      | $C3_{\kappa_F} = AC_1$ | $C4AC_1 / AC_2$      |
|-----------------------------|----------------|------------------------|----------------------|------------------------|----------------------|
| G1 — Humans                 | Script Norm.   | 0.898                  | 0.881 / 0.866        | 0.875                  | 0.751 / 0.774        |
|                             | Spelling       | 0.886                  | <b>1.000 / 1.000</b> | 0.898                  | 0.776 / 0.875        |
|                             | Grammatical    | 0.771                  | 0.864 / 0.843        | <b>1.000</b>           | <b>0.848 / 0.918</b> |
|                             | Code-Mixing    | <b>1.000</b>           | <b>1.000 / 1.000</b> | <b>1.000</b>           | 0.806 / 0.871        |
|                             | Correct        | 0.886                  | 0.886 / 0.886        | 0.886                  | 0.685 / 0.693        |
| G2 — Annotator LLMs         | Script Norm.   | −0.212                 | −0.222 / −0.225      | −0.250                 | 0.181 / 0.216        |
|                             | Spelling       | −0.111                 | −0.032 / 0.006       | −0.176                 | 0.262 / 0.350        |
|                             | Grammatical    | −0.081                 | 0.186 / 0.175        | −0.250                 | −0.013 / −0.012      |
|                             | Code-Mixing    | <b>−0.067</b>          | <b>0.219 / 0.304</b> | <b>−0.026</b>          | <b>0.149 / 0.160</b> |
|                             | Correct        | −0.176                 | 0.118 / 0.154        | −0.319                 | −0.070 / −0.057      |
| G3 — Non-Annotator LLMs     | Script Norm.   | 0.658                  | 0.254 / 0.378        | 0.792                  | 0.657 / 0.696        |
|                             | Spelling       | 0.693                  | 0.497 / 0.585        | <b>1.000</b>           | 0.202 / 0.242        |
|                             | Grammatical    | 0.202                  | <b>0.556 / 0.725</b> | 0.298                  | 0.353 / 0.392        |
|                             | Code-Mixing    | 0.286                  | 0.495 / 0.639        | 0.394                  | <b>0.486 / 0.595</b> |
|                             | Correct        | <b>0.762</b>           | −0.071 / −0.069      | 0.048                  | 0.551 / 0.623        |
| G5 — H + Annotator LLMs     | Script Norm.   | 0.048                  | 0.052 / 0.072        | 0.103                  | 0.072 / 0.038        |
|                             | Spelling       | 0.278                  | 0.086 / 0.098        | 0.153                  | −0.040 / −0.038      |
|                             | Grammatical    | 0.167                  | 0.129 / 0.139        | 0.236                  | 0.029 / 0.032        |
|                             | Code-Mixing    | 0.076                  | <b>0.446 / 0.506</b> | <b>0.350</b>           | <b>0.175 / 0.185</b> |
|                             | Correct        | <b>0.379</b>           | 0.445 / 0.487        | 0.236                  | 0.266 / 0.325        |
| G6 — H + Non-Annotator LLMs | Script Norm.   | 0.182                  | 0.235 / 0.271        | 0.385                  | 0.200 / 0.143        |
|                             | Spelling       | 0.609                  | 0.532 / 0.629        | 0.682                  | 0.251 / 0.266        |
|                             | Grammatical    | 0.235                  | 0.480 / 0.573        | <b>0.644</b>           | <b>0.361 / 0.395</b> |
|                             | Code-Mixing    | 0.323                  | <b>0.663 / 0.751</b> | 0.562                  | 0.430 / 0.494        |
|                             | Correct        | <b>0.670</b>           | 0.097 / 0.107        | 0.083                  | 0.458 / 0.513        |
| G7 — All Raters             | Script Norm.   | 0.101                  | 0.116 / 0.147        | 0.188                  | 0.176 / 0.145        |
|                             | Spelling       | 0.360                  | 0.206 / 0.243        | 0.358                  | 0.106 / 0.111        |
|                             | Grammatical    | 0.111                  | 0.245 / 0.291        | <b>0.331</b>           | 0.172 / 0.186        |
|                             | Code-Mixing    | 0.195                  | <b>0.493 / 0.570</b> | 0.432                  | <b>0.298 / 0.331</b> |
|                             | Correct        | <b>0.452</b>           | 0.167 / 0.189        | 0.080                  | 0.345 / 0.398        |

Table 3: Krippendorff’s  $\alpha$  (Nominal / Ordinal) by error category and rater group. Ordinal  $\alpha$  applies to C2 and C4 only.

| Rater Group                 | Error Category | C1Nominal     | C2Nom./Ord.          | C3Nominal    | C4Nom./Ord.          |
|-----------------------------|----------------|---------------|----------------------|--------------|----------------------|
| G1 — Humans                 | Script Norm.   | 0.900         | 0.884 / 0.867        | 0.878        | 0.757 / 0.765        |
|                             | Spelling       | 0.889         | <b>1.000 / 1.000</b> | 0.900        | 0.782 / 0.820        |
|                             | Grammatical    | 0.777         | 0.867 / 0.848        | <b>1.000</b> | <b>0.852 / 0.994</b> |
|                             | Code-Mixing    | <b>1.000</b>  | <b>1.000 / 1.000</b> | <b>1.000</b> | 0.811 / 0.869        |
|                             | Correct        | 0.889         | 0.889 / 0.889        | 0.889        | 0.693 / 0.689        |
| G2 — Annotator LLMs         | Script Norm.   | −0.182        | −0.191 / −0.192      | −0.219       | 0.201 / 0.289        |
|                             | Spelling       | −0.083        | −0.006 / 0.120       | −0.147       | 0.280 / 0.415        |
|                             | Grammatical    | −0.054        | 0.206 / 0.328        | −0.219       | 0.013 / 0.112        |
|                             | Code-Mixing    | <b>−0.040</b> | <b>0.239 / 0.396</b> | <b>0.000</b> | <b>0.280 / 0.415</b> |
|                             | Correct        | −0.147        | 0.140 / 0.257        | −0.286       | −0.044 / 0.011       |
| G3 — Non-Annotator LLMs     | Script Norm.   | 0.667         | 0.273 / 0.640        | 0.797        | 0.666 / 0.728        |
|                             | Spelling       | 0.701         | 0.509 / 0.666        | <b>1.000</b> | 0.222 / 0.296        |
|                             | Grammatical    | 0.222         | <b>0.568 / 0.828</b> | 0.316        | 0.370 / 0.440        |
|                             | Code-Mixing    | 0.304         | 0.507 / 0.763        | 0.409        | <b>0.499 / 0.679</b> |
|                             | Correct        | <b>0.768</b>  | −0.044 / −0.076      | 0.071        | 0.562 / 0.728        |
| G5 — H + Annotator LLMs     | Script Norm.   | 0.060         | 0.064 / 0.090        | 0.114        | 0.083 / 0.030        |
|                             | Spelling       | 0.287         | 0.097 / 0.118        | 0.163        | −0.027 / −0.026      |
|                             | Grammatical    | 0.177         | 0.140 / 0.153        | 0.246        | 0.042 / 0.046        |
|                             | Code-Mixing    | 0.088         | <b>0.453 / 0.549</b> | <b>0.358</b> | <b>0.186 / 0.201</b> |
|                             | Correct        | <b>0.386</b>  | 0.452 / 0.525        | 0.246        | 0.275 / 0.388        |
| G6 — H + Non-Annotator LLMs | Script Norm.   | 0.192         | 0.245 / 0.324        | 0.393        | 0.210 / 0.124        |
|                             | Spelling       | 0.614         | 0.538 / 0.699        | 0.686        | 0.260 / 0.289        |
|                             | Grammatical    | 0.245         | 0.486 / 0.580        | <b>0.649</b> | <b>0.369 / 0.428</b> |
|                             | Code-Mixing    | 0.331         | <b>0.667 / 0.818</b> | 0.568        | 0.437 / 0.545        |
|                             | Correct        | <b>0.674</b>  | 0.108 / 0.123        | 0.094        | 0.464 / 0.547        |
| G7 — All Raters             | Script Norm.   | 0.109         | 0.123 / 0.185        | 0.195        | 0.183 / 0.133        |
|                             | Spelling       | 0.365         | 0.212 / 0.277        | 0.364        | 0.113 / 0.121        |
|                             | Grammatical    | 0.118         | 0.252 / 0.315        | <b>0.336</b> | 0.179 / 0.200        |
|                             | Code-Mixing    | 0.202         | <b>0.498 / 0.621</b> | 0.437        | <b>0.304 / 0.358</b> |
|                             | Correct        | <b>0.456</b>  | 0.174 / 0.209        | 0.088        | 0.350 / 0.451        |

Table 4: Mean pairwise Cohen’s  $\kappa$  (Unweighted / Linear-Weighted) by error category and rater group. Weighted  $\kappa$  applies to ordinal C2 and C4 only.

| Rater Group                 | Error Category | C1Unwtd.     | C2Unwtd./Wtd.        | C3Unwtd.     | C4Unwtd./Wtd.        |
|-----------------------------|----------------|--------------|----------------------|--------------|----------------------|
| G1 — Humans                 | Script Norm.   | 0.898        | 0.882 / 0.867        | 0.875        | 0.752 / 0.774        |
|                             | Spelling       | 0.886        | <b>1.000 / 1.000</b> | 0.898        | 0.778 / 0.875        |
|                             | Grammatical    | 0.773        | 0.864 / 0.844        | <b>1.000</b> | <b>0.848 / 0.918</b> |
|                             | Code-Mixing    | <b>1.000</b> | <b>1.000 / 1.000</b> | <b>1.000</b> | 0.808 / 0.872        |
|                             | Correct        | 0.886        | 0.886 / 0.886        | 0.886        | 0.686 / 0.693        |
| G2 — Annotator LLMs         | Script Norm.   | −0.094       | −0.031 / 0.013       | 0.008        | 0.231 / 0.265        |
|                             | Spelling       | 0.000        | 0.082 / 0.145        | 0.000        | 0.296 / 0.386        |
|                             | Grammatical    | 0.000        | 0.259 / 0.268        | 0.000        | 0.104 / 0.113        |
|                             | Code-Mixing    | <b>0.048</b> | <b>0.286 / 0.375</b> | <b>0.100</b> | <b>0.296 / 0.386</b> |
|                             | Correct        | 0.000        | 0.167 / 0.207        | −0.034       | −0.006 / 0.016       |
| G3 — Non-Annotator LLMs     | Script Norm.   | 0.659        | 0.266 / 0.392        | 0.792        | 0.662 / 0.700        |
|                             | Spelling       | 0.694        | 0.500 / 0.588        | <b>1.000</b> | 0.262 / 0.299        |
|                             | Grammatical    | 0.222        | <b>0.560 / 0.727</b> | 0.300        | 0.358 / 0.394        |
|                             | Code-Mixing    | 0.286        | 0.520 / 0.650        | 0.400        | <b>0.489 / 0.596</b> |
|                             | Correct        | <b>0.762</b> | 0.134 / 0.152        | 0.200        | 0.556 / 0.625        |
| G5 — H + Annotator LLMs     | Script Norm.   | 0.069        | 0.151 / 0.166        | 0.195        | 0.212 / 0.206        |
|                             | Spelling       | 0.248        | 0.212 / 0.234        | 0.215        | 0.208 / 0.243        |
|                             | Grammatical    | 0.142        | 0.278 / 0.296        | 0.308        | 0.221 / 0.236        |
|                             | Code-Mixing    | 0.153        | <b>0.489 / 0.535</b> | <b>0.419</b> | <b>0.261 / 0.291</b> |
|                             | Correct        | <b>0.345</b> | 0.456 / 0.489        | 0.305        | 0.263 / 0.310        |
| G6 — H + Non-Annotator LLMs | Script Norm.   | 0.187        | 0.254 / 0.272        | 0.395        | 0.254 / 0.212        |
|                             | Spelling       | 0.613        | 0.554 / 0.635        | 0.683        | 0.351 / 0.386        |
|                             | Grammatical    | 0.297        | 0.500 / 0.588        | <b>0.649</b> | <b>0.403 / 0.440</b> |
|                             | Code-Mixing    | 0.391        | <b>0.687 / 0.752</b> | 0.576        | 0.446 / 0.513        |
|                             | Correct        | <b>0.669</b> | 0.205 / 0.225        | 0.194        | 0.465 / 0.516        |
| G7 — All Raters             | Script Norm.   | 0.117        | 0.163 / 0.198        | 0.231        | 0.235 / 0.216        |
|                             | Spelling       | 0.338        | 0.275 / 0.327        | 0.379        | 0.212 / 0.236        |
|                             | Grammatical    | 0.126        | 0.318 / 0.370        | <b>0.363</b> | 0.251 / 0.271        |
|                             | Code-Mixing    | 0.208        | <b>0.523 / 0.585</b> | 0.457        | <b>0.333 / 0.374</b> |
|                             | Correct        | <b>0.416</b> | 0.237 / 0.269        | 0.159        | 0.345 / 0.388        |

Table 5: ICC(A,1) Absolute / ICC(C,1) Consistency by error category and rater group (ordinal tasks C2 and C4 only; ICC is not defined for binary tasks).

| Rater Group                             | Error Category | C2ICC(A,1)/ICC(C,1)  | C4ICC(A,1)/ICC(C,1)  | C1         | C3         |
|-----------------------------------------|----------------|----------------------|----------------------|------------|------------|
| <i>(not applicable to binary tasks)</i> |                |                      |                      |            |            |
| G1 — Humans                             | Script Norm.   | 0.864 / 0.864        | 0.798 / 0.792        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | <b>1.000 / 1.000</b> | 0.936 / 0.936        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | 0.838 / 0.838        | <b>0.959 / 0.959</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | <b>1.000 / 1.000</b> | 0.927 / 0.930        | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.891 / 0.891        | 0.709 / 0.699        | <i>n/a</i> | <i>n/a</i> |
| G2 — Annotator LLMs                     | Script Norm.   | 0.046 / 0.079        | 0.296 / 0.327        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | 0.200 / 0.285        | 0.463 / 0.513        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | 0.286 / 0.372        | 0.123 / 0.158        | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | <b>0.448 / 0.564</b> | <b>0.463 / 0.513</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.243 / 0.270        | 0.036 / 0.040        | <i>n/a</i> | <i>n/a</i> |
| G3 — Non-Annotator LLMs                 | Script Norm.   | 0.526 / 0.543        | 0.735 / 0.741        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | 0.678 / 0.678        | 0.333 / 0.379        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | <b>0.851 / 0.857</b> | 0.427 / 0.415        | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | 0.752 / 0.770        | <b>0.677 / 0.666</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.170 / 0.286        | 0.678 / 0.671        | <i>n/a</i> | <i>n/a</i> |
| G5 — H + Annotator LLMs                 | Script Norm.   | 0.167 / 0.236        | 0.136 / 0.242        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | 0.207 / 0.339        | 0.134 / 0.356        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | 0.253 / 0.450        | 0.172 / 0.362        | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | <b>0.570 / 0.644</b> | <b>0.270 / 0.393</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.530 / 0.545        | 0.388 / 0.404        | <i>n/a</i> | <i>n/a</i> |
| G6 — H + Non-Annotator LLMs             | Script Norm.   | 0.314 / 0.309        | 0.147 / 0.167        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | 0.713 / 0.712        | 0.333 / 0.435        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | 0.653 / 0.679        | 0.442 / 0.481        | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | <b>0.812 / 0.815</b> | <b>0.559 / 0.585</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.185 / 0.251        | 0.561 / 0.550        | <i>n/a</i> | <i>n/a</i> |
| G7 — All Raters                         | Script Norm.   | 0.215 / 0.280        | 0.177 / 0.244        | <i>n/a</i> | <i>n/a</i> |
|                                         | Spelling       | 0.320 / 0.456        | 0.189 / 0.333        | <i>n/a</i> | <i>n/a</i> |
|                                         | Grammatical    | 0.369 / 0.524        | 0.251 / 0.379        | <i>n/a</i> | <i>n/a</i> |
|                                         | Code-Mixing    | <b>0.636 / 0.690</b> | <b>0.388 / 0.479</b> | <i>n/a</i> | <i>n/a</i> |
|                                         | Correct        | 0.246 / 0.324        | 0.449 / 0.457        | <i>n/a</i> | <i>n/a</i> |

Table 6: Mean Spearman  $\rho$  and Kendall  $\tau$  within rater group by error category. — = undefined: one rater in the pair assigned identical scores across all sentences in that category on a binary task, producing zero variance and making rank correlation mathematically undefined.

| Rater Group                 | Error Category | C1 $\rho$ / $\tau$   | C2 $\rho$ / $\tau$   | C3 $\rho$ / $\tau$   | C4 $\rho$ / $\tau$   |
|-----------------------------|----------------|----------------------|----------------------|----------------------|----------------------|
| G1 — Humans                 | Script Norm.   | 0.903 / 0.903        | 0.871 / 0.871        | 0.882 / 0.882        | 0.759 / 0.750        |
|                             | Spelling       | 0.892 / 0.892        | <b>1.000 / 1.000</b> | 0.903 / 0.903        | 0.838 / 0.824        |
|                             | Grammatical    | 0.794 / 0.794        | 0.853 / 0.852        | <b>1.000 / 1.000</b> | <b>0.995 / 0.977</b> |
|                             | Code-Mixing    | <b>1.000 / 1.000</b> | <b>1.000 / 1.000</b> | <b>1.000 / 1.000</b> | 0.884 / 0.867        |
|                             | Correct        | 0.892 / 0.892        | 0.892 / 0.892        | 0.892 / 0.892        | 0.686 / 0.675        |
| G2 — Annotator LLMs         | Script Norm.   | −0.150 / −0.150      | 0.191 / 0.183        | 0.015 / 0.015        | 0.350 / 0.345        |
|                             | Spelling       | —                    | 0.451 / 0.431        | —                    | 0.573 / 0.548        |
|                             | Grammatical    | —                    | <b>0.524 / 0.506</b> | —                    | 0.350 / 0.350        |
|                             | Code-Mixing    | <b>0.068 / 0.068</b> | 0.620 / 0.578        | <b>0.140 / 0.140</b> | <b>0.573 / 0.548</b> |
|                             | Correct        | —                    | 0.361 / 0.346        | −0.068 / −0.068      | 0.045 / 0.044        |
| G3 — Non-Annotator LLMs     | Script Norm.   | 0.663 / 0.663        | 0.686 / 0.602        | 0.792 / 0.792        | 0.738 / 0.715        |
|                             | Spelling       | 0.698 / 0.698        | 0.671 / 0.625        | <b>1.000 / 1.000</b> | 0.414 / 0.397        |
|                             | Grammatical    | 0.236 / 0.236        | <b>0.844 / 0.777</b> | 0.302 / 0.302        | 0.426 / 0.411        |
|                             | Code-Mixing    | 0.286 / 0.286        | 0.801 / 0.760        | 0.408 / 0.408        | <b>0.672 / 0.622</b> |
|                             | Correct        | <b>0.762 / 0.762</b> | 0.317 / 0.303        | 0.333 / 0.333        | 0.725 / 0.701        |
| G5 — H + Annotator LLMs     | Script Norm.   | −0.038 / −0.038      | 0.130 / 0.125        | 0.139 / 0.139        | 0.048 / 0.048        |
|                             | Spelling       | 0.319 / 0.319        | 0.187 / 0.180        | 0.233 / 0.233        | 0.184 / 0.178        |
|                             | Grammatical    | 0.038 / 0.038        | 0.369 / 0.354        | 0.458 / 0.458        | 0.220 / 0.218        |
|                             | Code-Mixing    | −0.022 / −0.022      | <b>0.577 / 0.557</b> | <b>0.374 / 0.374</b> | <b>0.323 / 0.308</b> |
|                             | Correct        | <b>0.594 / 0.594</b> | 0.526 / 0.514        | 0.306 / 0.306        | 0.409 / 0.393        |
| G6 — H + Non-Annotator LLMs | Script Norm.   | −0.111 / −0.111      | 0.113 / 0.110        | 0.185 / 0.185        | −0.148 / −0.143      |
|                             | Spelling       | 0.539 / 0.539        | 0.653 / 0.613        | 0.557 / 0.557        | 0.408 / 0.398        |
|                             | Grammatical    | 0.223 / 0.223        | 0.519 / 0.485        | <b>0.655 / 0.655</b> | 0.435 / 0.421        |
|                             | Code-Mixing    | 0.327 / 0.327        | <b>0.793 / 0.773</b> | 0.562 / 0.562        | <b>0.496 / 0.466</b> |
|                             | Correct        | <b>0.594 / 0.594</b> | −0.046 / −0.054      | −0.108 / −0.108      | 0.466 / 0.447        |
| G7 — All Raters             | Script Norm.   | 0.147 / 0.147        | 0.324 / 0.305        | 0.255 / 0.255        | 0.223 / 0.217        |
|                             | Spelling       | 0.529 / 0.529        | 0.459 / 0.436        | 0.591 / 0.591        | 0.351 / 0.340        |
|                             | Grammatical    | 0.202 / 0.202        | 0.503 / 0.477        | 0.558 / 0.558        | 0.388 / 0.380        |
|                             | Code-Mixing    | 0.229 / 0.229        | <b>0.700 / 0.675</b> | 0.490 / 0.490        | <b>0.465 / 0.444</b> |
|                             | Correct        | <b>0.626 / 0.626</b> | 0.299 / 0.287        | 0.164 / 0.164        | 0.468 / 0.453        |

Table 7: Exact agreement (%) by error category and rater group. No chance correction is applied; this is the raw fraction of rater pairs that assigned identical scores.

| Rater Group                 | Error Category | C1 Det.      | C2 Span      | C3 Cat.      | C4 Corr.    |
|-----------------------------|----------------|--------------|--------------|--------------|-------------|
| G1 — Humans                 | Script Norm.   | 95.0         | 95.0         | 95.0         | 90.0        |
|                             | Spelling       | 95.0         | <b>100.0</b> | 95.0         | 95.0        |
|                             | Grammatical    | 95.0         | 95.0         | <b>100.0</b> | 95.0        |
|                             | Code-Mixing    | <b>100.0</b> | <b>100.0</b> | <b>100.0</b> | 90.0        |
|                             | Correct        | 95.0         | 95.0         | 95.0         | 85.0        |
| G2 — Annotator LLMs         | Script Norm.   | 65.0         | 25.0         | 40.0         | 70.0        |
|                             | Spelling       | 80.0         | 55.0         | 70.0         | 75.0        |
|                             | Grammatical    | <b>85.0</b>  | 60.0         | 60.0         | 70.0        |
|                             | Code-Mixing    | 60.0         | <b>55.0</b>  | <b>55.0</b>  | <b>65.0</b> |
|                             | Correct        | 70.0         | 60.0         | 40.0         | 60.0        |
| G3 — Non-Annotator LLMs     | Script Norm.   | 85.0         | 60.0         | 90.0         | 80.0        |
|                             | Spelling       | 85.0         | 70.0         | <b>100.0</b> | 55.0        |
|                             | Grammatical    | 65.0         | 75.0         | 65.0         | 65.0        |
|                             | Code-Mixing    | 70.0         | <b>70.0</b>  | 70.0         | <b>70.0</b> |
|                             | Correct        | <b>90.0</b>  | 45.0         | 60.0         | 80.0        |
| G5 — H + Annotator LLMs     | Script Norm.   | 30.0         | 10.0         | 20.0         | 15.0        |
|                             | Spelling       | 60.0         | 15.0         | 35.0         | 15.0        |
|                             | Grammatical    | 75.0         | 15.0         | 35.0         | 15.0        |
|                             | Code-Mixing    | 50.0         | <b>45.0</b>  | <b>40.0</b>  | <b>25.0</b> |
|                             | Correct        | <b>60.0</b>  | 55.0         | 35.0         | 50.0        |
| G6 — H + Non-Annotator LLMs | Script Norm.   | 40.0         | 40.0         | 55.0         | 35.0        |
|                             | Spelling       | 70.0         | 55.0         | 75.0         | 45.0        |
|                             | Grammatical    | 50.0         | 60.0         | 65.0         | 50.0        |
|                             | Code-Mixing    | 60.0         | <b>65.0</b>  | 60.0         | <b>55.0</b> |
|                             | Correct        | <b>75.0</b>  | 20.0         | 20.0         | 60.0        |
| G7 — All Raters             | Script Norm.   | 20.0         | 5.0          | 20.0         | 10.0        |
|                             | Spelling       | 50.0         | 10.0         | 35.0         | 10.0        |
|                             | Grammatical    | 40.0         | 15.0         | 25.0         | 10.0        |
|                             | Code-Mixing    | <b>45.0</b>  | <b>30.0</b>  | <b>40.0</b>  | <b>25.0</b> |
|                             | Correct        | 50.0         | 5.0          | 5.0          | 45.0        |